EN IEC 55014-1:2021
EN IEC 55014-2:2021
EN IEC 61000-3-2:2019+A2:2024
EN 61000-3-3:2013+A2:2021+AC:2022-01

TEST REPORT

For

Quanzhou STAYTRUE Electronic Technology Co., Ltd

Unit 1A-2, 1st Floor, Building 1, Fengshu Logistics Park, No. 920, Binhu South Road, Taiwan Investment Zone, Quanzhou, Fujian Province, China

Tested Model: SDC-LE-01D-4pin

Report Type:	**Product Name:**
Original Report	Module SDC-LE-01D

Report Number:	2607S04032E-EM-01
Report Date:	2026-05-06
Reviewed By:	Ash Lin
Approved By:	Miles Chen
Prepared By:	Bay Area Compliance Laboratories Corp.(Fujian) 4/F., No.65, Siming Industrial Zone, Meixi Road, Tong'an District, Xiamen, Fujian, China Tel: +86-592-3200111 www.baclcorp.com.cn

TABLE OF CONTENTS

REPORT REVISION HISTORY **4**

GENERAL INFORMATION **5**

Product Description for Equipment under Test (EUT) 5
Objective 5
Test Facility 5
Measurement Uncertainty 6

SYSTEM TEST CONFIGURATION **7**

Test Mode and Voltage 7
EUT Exercise Software 7
Special Accessories 7
Equipment Modifications 7
Support Equipment List and Details 7
External I/O Cable 7
Block Diagram of Test Setup 8

SUMMARY OF TEST REPORT **9**

TEST EQUIPMENT LIST **10**

EN IEC 55014-1 §4.3.3 DISTURBANCE VOLTAGE **11**

Test System Setup 11
EMI Test Receiver Setup 11
Test Procedure 11
Result &Margin Calculation 12
Test Data 12

EN IEC 55014-1 §4.4 DISCONTINUOUS DISTURBANCE **13**

Test System Setup 13
Test Procedure 13
Test Data 13

EN IEC 55014-1 §4.3.4 RADIATED DISTURBANCES (30 MHz to 1 GHz) **14**

Test System Setup 14
EMI Test Receiver Setup 15
Test Procedure 15
Result & Margin Calculation 15
Test Data 16

EN IEC 55014-1 §4.3.5 RADIATED DISTURBANCES (Above 1 GHz) **18**

Test System Setup 18
EMI Test Receiver Setup 18
Test Procedure 18
Result & Margin Calculation 19
Test Data 19

EN IEC 61000-3-2 HARMONIC CURRENT EMISSIONS **20**

Test System Setup 20
Test Standard 20
Test procedure and Classification of equipment 20
Application of limits: 21
Test Data 23

EN 61000-3-3 VOLTAGE FLUCTUATIONS AND FLICKER **24**

Test System Setup 24
Test Standard 24
Test procedure 24
Flicker Test Limits 24

Test Data 25

EN IEC 55014-2 §5.1 ELECTROSTATIC DISCHARGE (IEC 61000-4-2) 26

Test System Setup 26
Test Specification 26
Test Procedure 27
Test Data 28

EN IEC 55014-2 §5.5 RADIO FREQUENCY ELECTROMAGNETIC FIELDS (IEC 61000-4-3) 29

Test System Setup 29
Test specification 29
Test Procedure 29
Test Data 30

EN IEC 55014-2 §5.2 ELECTRIC FAST TRANSIENTS (IEC 61000-4-4) 31

Test System Setup 31
Test specification 31
Test Procedure 31
Test Data 31

EN IEC 55014-2 §5.6 SURGES (IEC 61000-4-5) 32

Test System Setup 32
Test Specification 32
Test Procedure 32
Test Data 32

EN IEC 55014-2 §5.4 INJECTED CURRENTS (IEC 61000-4-6) 33

Test System Setup 33
Test specification 33
Test Procedure 33
Test Data 34

EN IEC 55014-2 §5.7 VOLTAGE DIPS AND INTERRUPTIONS (IEC 61000-4-11) 35

Test System Setup 35
Test specification 35
Test Procedure 35
Test Data 35

EXHIBIT A - EUT PHOTOGRAPHS 36

EXHIBIT B - TEST SETUP PHOTOGRAPHS 41

PRODUCT SIMILARITY DECLARATION LETTER 45

REPORT REVISION HISTORY

Number of Revisions	Report No.	Version	Issue Date	Description
0	2607S04032E-EM-01	R1V1	2026-05-06	Initial Release

GENERAL INFORMATION

Product Description for Equipment under Test (EUT)

Applicant:	Quanzhou STAYTRUE Electronic Technology Co., Ltd
Applicant Address:	Unit 1A-2, 1st Floor, Building 1, Fengshu Logistics Park, No. 920, Binhu South Road, Taiwan Investment Zone, Quanzhou, Fujian Province, China
Manufacturer:	Quanzhou STAYTRUE Electronic Technology Co., Ltd
Manufacturer Address:	Unit 1A-2, 1st Floor, Building 1, Fengshu Logistics Park, No. 920, Binhu South Road, Taiwan Investment Zone, Quanzhou, Fujian Province, China
Product Name:	Module SDC-LE-01D
Tested Model:	SDC-LE-01D-4pin
Multiple Model(s):	SDC-LE-01D-5pin
Trade mark:	SDC-LE-01D
Power Supply:	DC 3.3V
★Highest Clock Frequency:	48 MHz
EUT Received Status:	Good
Note: *1. The highest clock frequency is provided by the applicant.* *2. The test model is identify with the series model except for the model name and number of pins, please refer to declaration letter for more detail.* *3. All measurement and test data in this report was gathered from production sample serial number: 3JM8-1 (Assigned by the BACL (Fujian). The EUT supplied by the applicant was received on 2026-04-08).*	

Objective

The test report is prepared for *Quanzhou STAYTRUE Electronic Technology Co., Ltd* in accordance with

EN IEC 55014-1:2021, Electromagnetic compatibility - Requirements for household appliances, electric tools and similar apparatus - Part 1: Emission.

EN IEC 55014-2:2021, Electromagnetic compatibility - Requirements for household appliances, electric tools and similar apparatus - Part 2: Immunity - Product family standard.

EN IEC 61000-3-2:2019+A2:2024, Electromagnetic compatibility (EMC) - Part 3-2: Limits - Limits for harmonic current emissions (equipment input current ≤16 A per phase).

EN 61000-3-3:2013+A2:2021+AC:2022-01, Electromagnetic compatibility (EMC) - Part 3-3: Limits - Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems, for equipment with rated current ≤16 A per phase and not subject to conditional connection.

The objective of the manufacturer is to determine the compliance of the EUT with EN IEC 55014-1, EN IEC 55014-2, EN IEC 61000-3-2 and EN 61000-3-3.

Test Facility

The Test site used by Bay Area Compliance Laboratories Corp. (Fujian) to collect test data is located on the Unit 302, No. 902, Meifeng South Road, Tong'an District, Xiamen, Fujian, China

Measurement Uncertainty

Where relevant, the following measurement uncertainty levels have been estimated for tests performed on the product as specified in CISPR 16-4-2. This uncertainty represents expanded uncertainty expressed at 95% confidence level using a coverage factor of k=2.
If U_{lab} is less than or equal to U_{cispr}, then compliance is deemed to occur if no measured disturbance level exceeds the disturbance limit.

$$u_c(y) = \sqrt{\sum_i c_i^2 \; u^2(x_i)}$$

Item	Frequency Range	U_{cispr}	Ulab= 2 uc (y) (Confidence of 95%)
Radiated Emissions	30MHz~200MHz (Horizontal)	5.11dB	3.33dB
	30MHz~200MHz (Vertical)	5.26dB	3.51dB
	200MHz~1GHz (Horizontal)	5.11dB	4.87dB
	200MHz~1GHz (Vertical)	5.26dB	4.99dB

SYSTEM TEST CONFIGURATION

Test Mode and Voltage

The system was configured for testing in a typical mode (as normally used by a typical user).	
Test mode:	Test Mode 1: Operation + BT Link
Test voltage:	DC 3.3V from serial interface board
Remark:	During all emission tests, the EUT was configured to measure its highest possible emission level and the worst case's test data was presented in this test report.

EUT Exercise Software

No exercise software was used to test.

Special Accessories

No special accessory was used.

Equipment Modifications

No modification was made to the EUT tested.

Support Equipment List and Details

Manufacturer	Description	Model	Serial Number
Xiaomi	Mobile phone	23078RKD5C	48435/63ZN02677
Lenovo	Laptop	Lenovo 14w Gen 2	S43111908001
Unknown	serial interface board	Unknown	Unknown

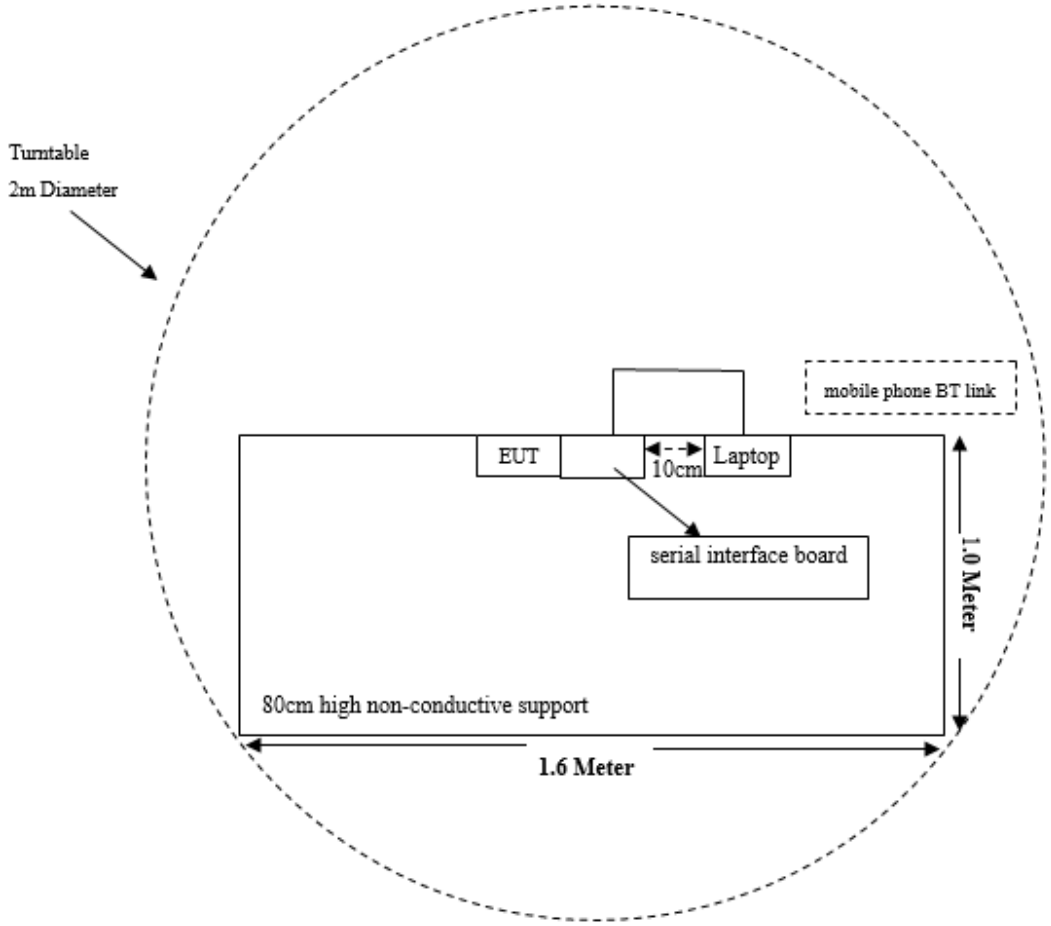
External I/O Cable

Cable Description	Length (m)	From	To
Type-C Cable	1	EUT	Laptop

Block Diagram of Test Setup

Radiated disturbance(30MHz-1000MHz):

Test Mode 1:



SUMMARY OF TEST REPORT

EN IEC 55014-1		
Clause	**Description**	**Results**
§4.3.3	Disturbance voltage	Not Applicable (See Note 1)
§4.4	Discontinuous disturbance	Not Applicable (See Note 1)
§4.3.4	Radiated disturbances (30 MHz to 1 000 MHz)	Compliant
§4.3.5	Radiated disturbances (Above1 GHz)	Not Applicable (See Note 2)
EN IEC 61000-3-2		
Clause	**Description**	**Results**
§6, §7	Harmonic current Emissions	Not Applicable (See Note 1)
EN 61000-3-3		
Clause	**Description**	**Results**
§5, §6	Voltage fluctuation and flicker	Not Applicable (See Note 1)
EN IEC 55014-2(Category IV)		
Clause	**Description**	**Results** (See Note 3)
§5.1	Electrostatic discharge (IEC 61000-4-2)	Compliant
§5.5	Radio frequency electromagnetic fields (IEC 61000-4-3)	Compliant
§5.2	Electric fast transients (IEC 61000-4-4)	Not Applicable (See Note 1)
§5.6	Surges (IEC 61000-4-5)	Not Applicable (See Note 1)
§5.4	Injected currents (0.15MHz to 80MHz) (IEC 61000-4-6)	Not Applicable (See Note 1)
§5.7	Voltage dips and interruptions (IEC 61000-4-11)	Not Applicable (See Note 1)

Notes:

1. The EUT is powered by DC source.
2. The item applies only to equipment with highest clock frequency greater than 108 MHz.
3. Immunity test performance criteria refer to EN IEC 55014-2 §6.

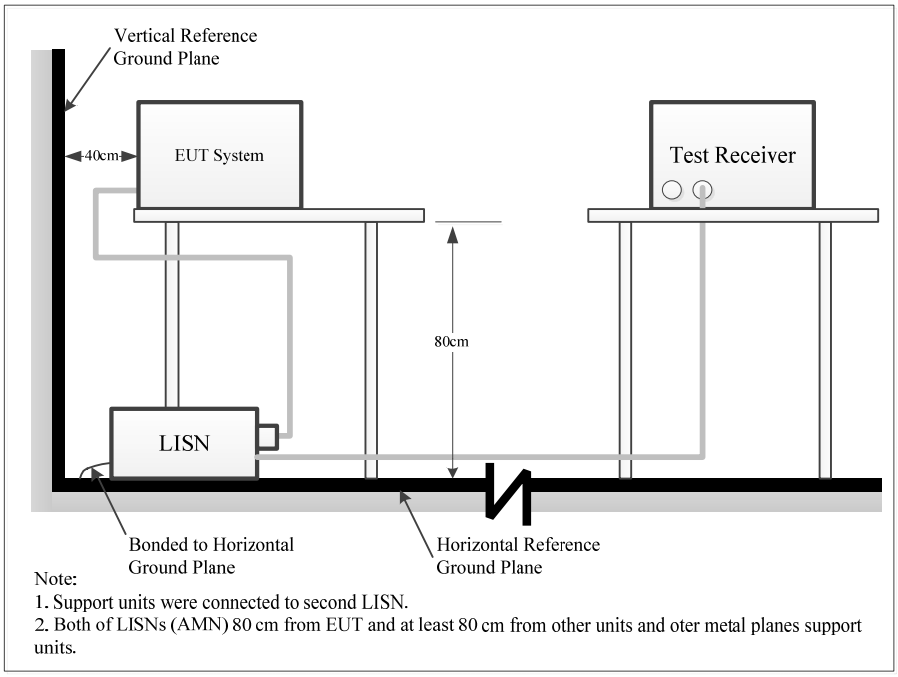
TEST EQUIPMENT LIST

Test Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due Date
Radiated Emissions 30 MHz to 1 GHz					
EMI Test Receiver	Rohde & Schwarz	ESR3	103103	2026/01/21	2027/01/20
Antenna	Sunol Sciences	JB6	A122022-5	2023/07/27	2026/07/26
Amplifier	Sonoma	310B	120903	2026/01/22	2027/01/21
Coaxial Cable	XINHANGWEIBO	XH400T-N-4M	CC002	2026/01/20	2027/01/19
Coaxial Cable	XINHANGWEIBO	XH460B-N-2M	CC006	2026/01/20	2027/01/19
Coaxial Cable	XINHANGWEIBO	XH460B-N-12M	CC007	2026/01/20	2027/01/19
Test Software	Audix	E3	18621a	N/A	N/A
Electrostatic discharge (ESD)					
ESD Simulator	TESEQ	NSG 438	724	2026/01/22	2027/01/21
Test Software	TESEQ	Pistol	V 01.00	N/A	N/A
Continuous RF electromagnetic field disturbances					
Log Periodic Antenna	Amplifier Research	ATL80M1G	583474	N/A	N/A
RF Power Amplifier	Amplifier Research	150W1000MS	303367	2026/01/22	2027/01/21
High Power Directional Coupler	WERLATONE	06934	11947	2026/01/21	2027/01/20
Microwave Analog Signal Generator	Agilent	N5181A	MY48180319	2026/01/21	2027/01/20
Power meter	Agilent	E4419B	MY45100315	2025/09/25	2026/09/24
Power Sensor	Agilent	E9301A	MY52010003	2026/01/22	2027/01/21
Test Software	BACL	VEE PRO	V 2.7.6	N/A	N/A

Statement of Traceability: Bay Area Compliance Laboratories Corp. (Fujian) attests that all calibrations have been performed in accordance to requirements that traceable to National Primary Standards and International System of Units (SI).

EN IEC 55014-1 §4.3.3 DISTURBANCE VOLTAGE

Test System Setup



Note:
1. Support units were connected to second LISN.
2. Both of LISNs (AMN) 80 cm from EUT and at least 80 cm from other units and oter metal planes support units.

The setup of EUT is according with CISPR 16-2-1:2014+COR1:2020 measurement procedures. The specification used was with the EN IEC 55014-1: 2021 limits.

The external I/O cables were draped along the test table and formed a bundle 30 to 40 cm long in the middle.

The spacing between the peripherals was 10 cm.

EMI Test Receiver Setup

The EMI test receiver was set to investigate the spectrum from 150 kHz to 30 MHz

During the conducted emission test, the EMI test receiver was set with the following configurations:

Frequency Range	IF B/W
150 kHz - 30 MHz	9 kHz

Test Procedure

During the conducted emission test, the EUT was connected to the outlet of the first LISN.

Maximizing procedure was performed on the six (6) highest emissions of the EUT.

All data was recorded in the Quasi-peak and average detection mode. If the maximum peak value of the emissions is below the average limit, the QP value and average value measurement will not need to be performed and only record the maximum peak measured value to meet the requirements.

Result &Margin Calculation

The Result is calculated by adding LISN VDF (Voltage Division Factor), Cable Loss and Transient Limiter Attenuation from the Meter Reading. The basic equation is as follows:

Factor (dB) = LISN VDF (dB) + Cable Loss (dB) + Transient Limiter Attenuation (dB)
Result (dBµV) = Reading (dBµV) + Factor (dB)

The "**Margin**" column of the following data tables indicates the degree of compliance with the applicable limit. For example, a margin of 7dB means the emission is 7dB below the limit. The equation for margin calculation is as follows:

Margin (dB) = Limit (dBµV) – Result (dBµV)

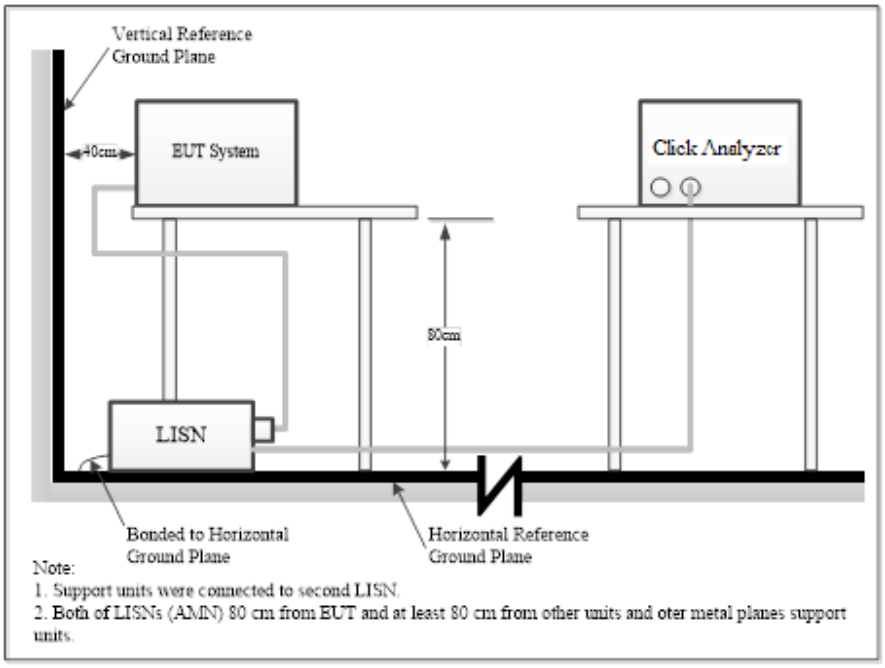
Test Data

Test Result: N/A.

Note: The EUT is powered by DC source.

EN IEC 55014-1 §4.4 DISCONTINUOUS DISTURBANCE

Test System Setup



The setup of EUT is according with CISPR 16-2-1:2014+COR1:2020 measurement procedures. The specification used was with the EN IEC 55014-1: 2021 limits.

The external I/O cables were draped along the test table and formed a bundle 30 to 40 cm long in the middle.

The spacing between the peripherals was 10 cm.

Test Procedure

This test is done when switch operations in thermostatically controlled appliances, automatic program controlled machines and other electrically controlled or operated appliances may generate discontinuous disturbance (Click). The measurement of disturbance shall be performed at the following restricted number of frequencies: 150kHz, 500kHz, 1.4MHz and 30MHz. At each frequency, for appliances, which stop automatically, duration of the minimum number of complete programs necessary to produce 40 counted clicks or, where relevant, 40 counted clicks have not been produced, the test is stopped at the end of the program in course. The relevant click rate N. The appliance under test shall be deemed to comply with the limit if not more than a quarter of the number of the counted click registered during the observation time.

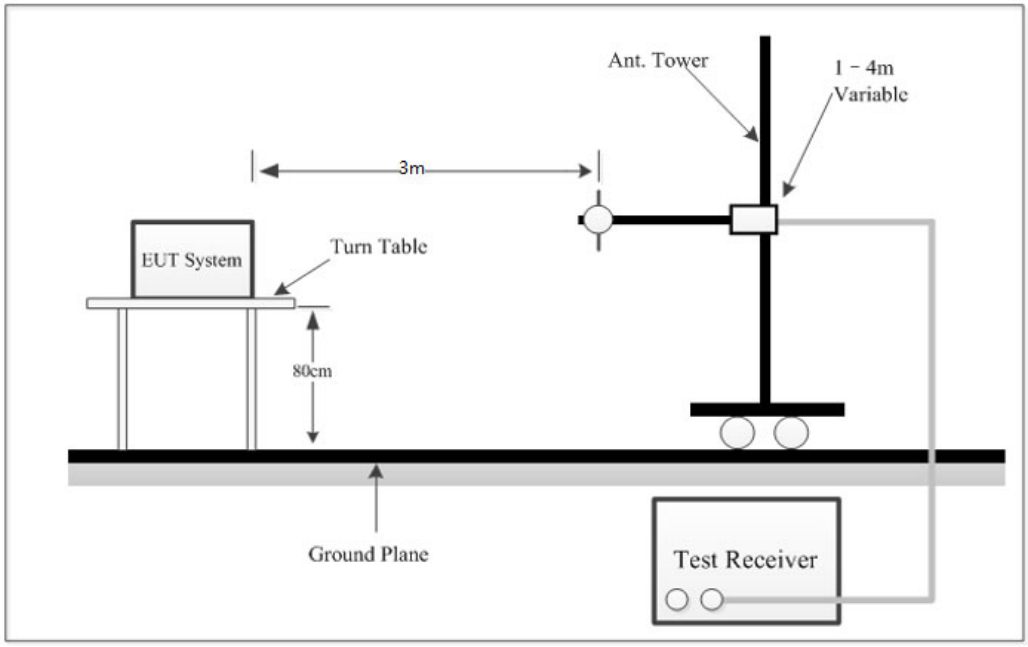
Test Data

Test Result: N/A.

Note: The EUT is powered by DC source.

EN IEC 55014-1 §4.3.4 RADIATED DISTURBANCES (30 MHz to 1 GHz)

Test System Setup



Radiated Top View:

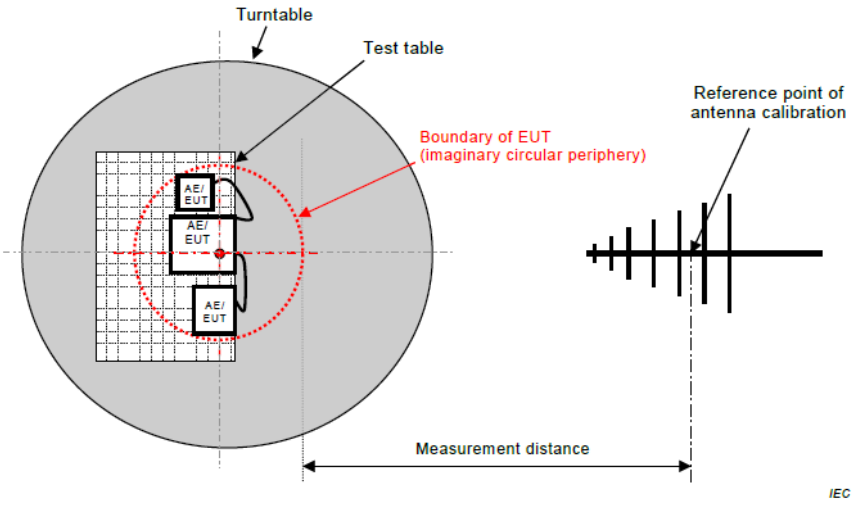


Figure C.1 – Measurement distance

The radiated emission tests were performed in the 3 meters chamber test site, using the setup accordance with CISPR 16-1-4:2019+A1:2020, CISPR 16-2-3:2016+A1:2019. The specification used was EN IEC 55014-1:2021 limits.

The external I/O cables were draped along the test table and formed a bundle 30 to 40 cm long in the middle.

The spacing between the peripherals was 10 cm.

EMI Test Receiver Setup

The system was investigated from 30 MHz to 1 GHz.

During the radiated emission test, the EMI test receiver was set with the following configurations:

Frequency Range	RBW	VBW	Measurement	Detector
30 MHz – 1000 MHz	100 kHz	300 kHz	PK	PK
	120 kHz	/	QP	QP

Test Procedure

Maximizing procedure was performed on the six (6) highest emissions to ensure that the EUT complied with all installation combinations.

If the measured peak level of the emissions that the measuring receiver reading level plus corrected factor is at least 10 dB below the QP emission limit, there's no need to record the measured QP level of the emissions in the report.

Result & Margin Calculation

The Result is calculated by adding the Antenna Factor and Cable Loss, and subtracting the Amplifier Gain from the Meter Reading. The basic equation is as follows:

Result (dBμV/m) = Reading (dBμV) + Antenna Factor (dB/m) + Cable Loss (dB) - Amplifier Gain (dB)

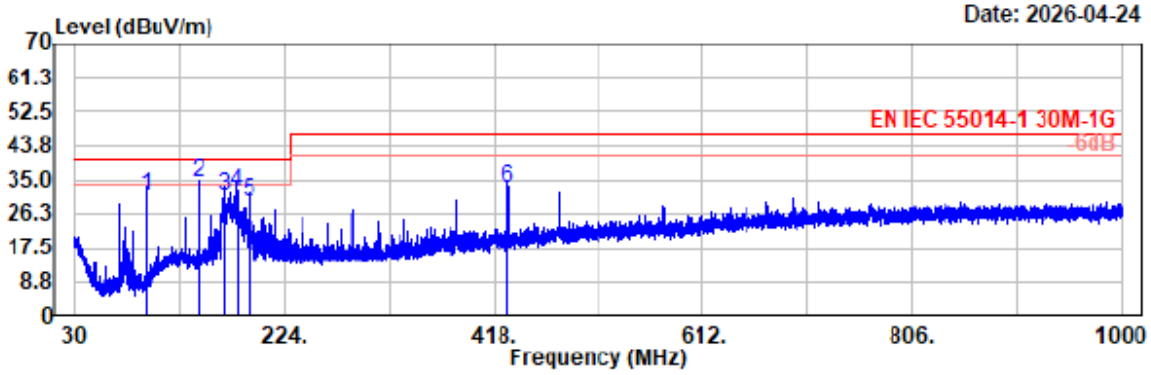
The "**Margin**" column of the following data tables indicates the degree of compliance with the applicable limit. For example, a margin of 7dB means the emission is 7dB below the limit. The equation for margin calculation is as follows:

Margin (dB) = Limit (dBμV/m) – Result (dBμV/m)

Test Data

Project No.: 2607S04032E-EM
Test Mode: Mode1
EUT Model: SDC-LE-01D-4pin
Test distance: 3m

Temp/Humi/ATM: 22.8℃/66%/100.1kPa
Tested by: Zane Zhang
Power Source: DC 3.3V

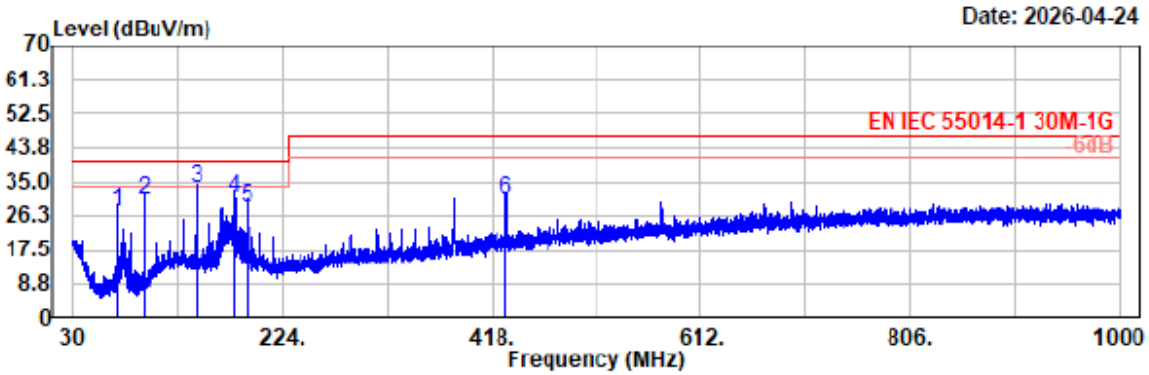


Condition: PK RBW:100kHz VBW:300kHz SWT:auto

Freq	Reading	Factor	Result	Limit	Margin	Polarity	Remark
MHz	dBuV	dB/m	dBuV/m	dBuV/m	dB		
95.86	47.24	-16.14	31.10	40.00	8.90	Horizontal	QP
143.87	45.23	-10.94	34.29	40.00	5.71	Horizontal	QP
167.84	42.75	-11.67	31.08	40.00	8.92	Horizontal	QP
179.90	44.30	-12.34	31.96	40.00	8.04	Horizontal	QP
191.70	41.82	-12.17	29.65	40.00	10.35	Horizontal	QP
431.48	38.30	-5.35	32.95	47.00	14.05	Horizontal	QP

Project No.: 2607S04032E-EM
Test Mode: Mode1
EUT Model: SDC-LE-01D-4pin
Test distance: 3m

Temp/Humi/ATM: 22.8℃/66%/100.1kPa
Tested by: Zane Zhang
Power Source: DC 3.3V

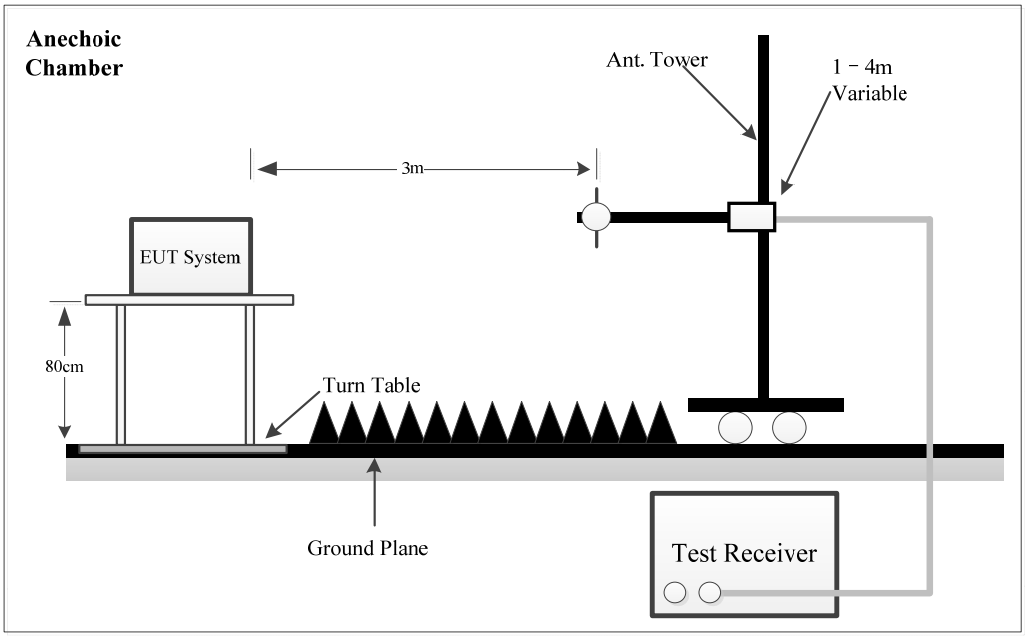


Condition: PK RBW:100kHz VBW:300kHz SWT:auto

Freq	Reading	Factor	Result	Limit	Margin	Polarity	Remark
MHz	dBuV	dB/m	dBuV/m	dBuV/m	dB		
71.90	44.67	-17.02	27.65	40.00	12.35	Vertical	QP
95.86	46.68	-16.14	30.54	40.00	9.46	Vertical	QP
143.90	44.28	-10.94	33.34	40.00	6.66	Vertical	QP
179.77	43.36	-12.33	31.03	40.00	8.97	Vertical	QP
191.80	40.84	-12.17	28.67	40.00	11.33	Vertical	QP
431.58	35.76	-5.35	30.41	47.00	16.59	Vertical	QP

EN IEC 55014-1 §4.3.5 RADIATED DISTURBANCES (Above 1 GHz)

Test System Setup



The radiated emission tests were performed in the 3 meters, using the setup accordance with the CISPR 16-1-4:2019+A1:2020, CISPR 16-2-3:2016+A1:2019. The specification used was EN IEC 55014-1:2021 limits.

The external I/O cables were draped along the test table and formed a bundle 30 to 40 cm long in the middle.

The spacing between the peripherals was 10 cm.

EMI Test Receiver Setup

The system was investigated from above 1 GHz.

During the radiated emission test, the EMI test receiver was set with the following configurations:

Frequency Range	RBW	VBW	Measurement	Detector
Above 1 GHz	1 MHz	3 MHz	PK	PK
	1 MHz	10 Hz	AV	PK

Test Procedure

Maximizing procedure was performed on the six (6) highest emissions to ensure that the EUT complied with all installation combinations.

If the maximum peak value of the emissions is below the average limit, the average value measurement will not need to be performed and only record the maximum peak measured value to meet the requirements.

Result & Margin Calculation

The Result is calculated by adding the Antenna Factor and Cable Loss, and subtracting the Amplifier Gain from the Meter Reading. The basic equation is as follows:

Result (dBµV/m) = Reading (dBµV) + Antenna Factor (dB/m) + Cable Loss (dB) - Amplifier Gain (dB)

The "**Margin**" column of the following data tables indicates the degree of compliance with the applicable limit. For example, a margin of 7dB means the emission is 7dB below the limit. The equation for margin calculation is as follows:

Margin (dB) = Limit (dBµV/m) – Result (dBµV/m)

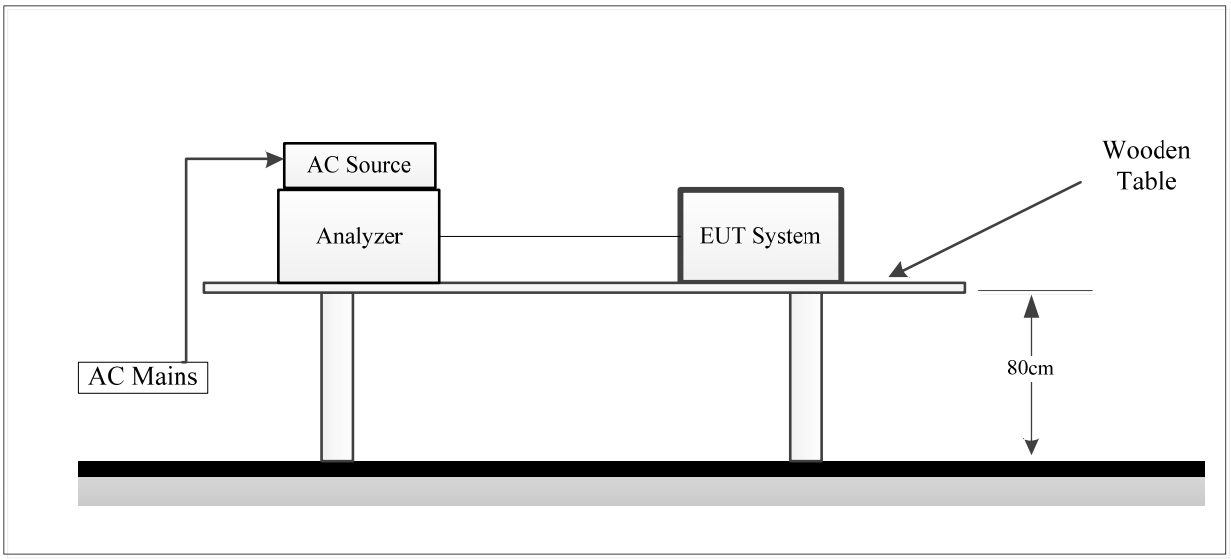
Test Data

Test Result: N/A.

Note: The item applies only to equipment with highest clock frequency greater than 108 MHz.

EN IEC 61000-3-2 HARMONIC CURRENT EMISSIONS

Test System Setup



Test Standard

EN IEC 61000-3-2:2019+A2:2024

Test procedure and Classification of equipment

The EUT was placed on the top of a wooden table 0.8 meters above the ground and operated to produce the maximum harmonic components under Mode 1conditions for each successive harmonic component in turn. The classification of EUT is according to section 5 of EN IEC 61000-3-2.

The EUT is classified as follows:

Class A:
Equipment not specified as belonging to Class B, C or D shall be considered as Class A equipment.
Some examples of Class A equipment are:
- balanced three-phase equipment;
- household appliances, excluding those specified as belonging to Class B, C or D;
- vacuum cleaners;
- high pressure cleaners;
- tools, excluding portable tools;
- independent phase control dimmers;
- audio equipment;
- professional luminaires for stage lighting and studios.

Class B:
- portable tools;
- arc welding equipment which is not professional equipment.

Class C:
- lighting equipment.

Class D:
Equipment having a specified power less than or equal to 600 W, of the following type:

- personal computer and personal computer monitors;
- television receivers;
- refrigerators and freezers having one or more variable-speed drives to control compressor motor(s).

Application of limits:

Limits for Class A equipment:
The harmonics of the input current shall not exceed the values given in Table 1.

Table 1 – Limits for Class A equipment

Harmonic order h	Maximum permissible harmonic current A
Odd harmonics	
3	2,30
5	1,14
7	0,77
9	0,40
11	0,33
13	0,21
$15 \le h \le 39$	$0{,}15\frac{15}{h}$
Even harmonics	
2	1,08
4	0,43
6	0,30
$8 \le h \le 40$	$0{,}23\frac{8}{h}$

Limits for Class B equipment:
For Class B equipment, the harmonics of the input current shall not exceed the values given in Table 1 multiplied by a factor of 1.5.

Limits for Class C equipment:
Rated power > 25 W

For luminaires with incandescent lamps and built-in phase control dimming having a rated power greater than 25 W, the harmonics of the input current shall not exceed the limits given in Table 1.

For any other lighting equipment having a rated power greater than 25 W, the harmonics of the input current shall not exceed the relative limits given in Table 2. For those types that include means for control (e.g. dimming, colour), the harmonics of the input current shall not exceed the harmonic current values derived from the percentage limits given in Table 2 for the maximum active input power (P_{max}) condition when tested in both following conditions:

– with the means for control set to obtain P_{max};
– with the means for control set to the position expected to produce the maximum total harmonic current (THC) within the active input power range [P_{min}, P_{max}], where
- P_{min} = 5 W, if $P_{max} \le 50$ W;
- P_{min} = 10 % of P_{max}, if 50 W < $P_{max} \le 250$ W;
- P_{min} = 25 W, if P_{max} > 250 W.

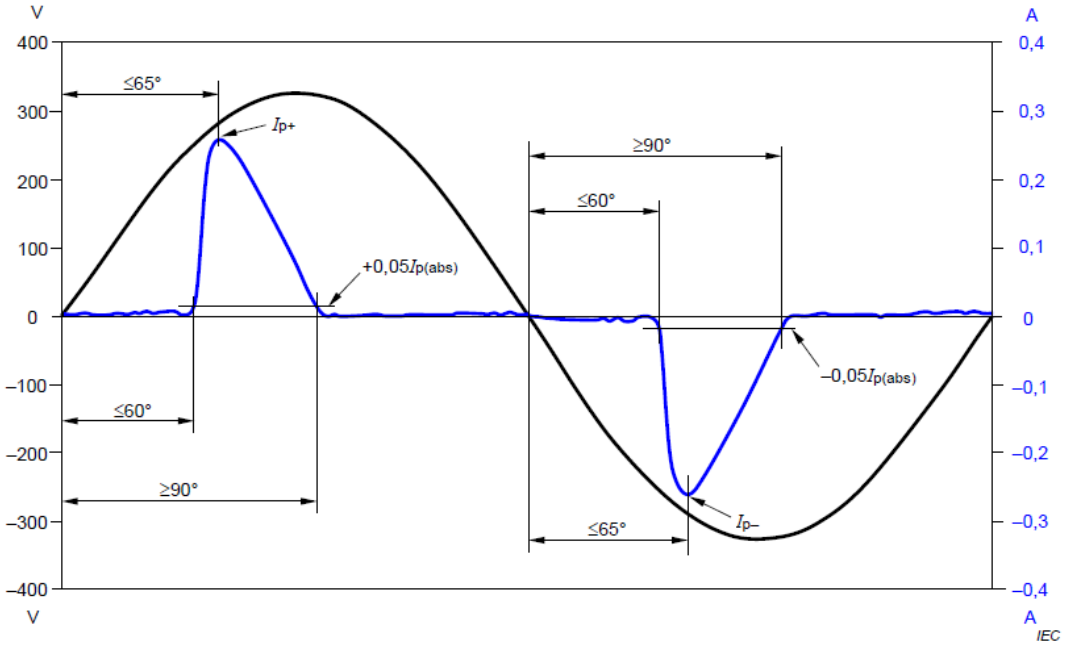
Table 2 – Limits for Class C equipment [a]

Harmonic order h	Maximum permissible harmonic current expressed as a percentage of the input current at the fundamental frequency %
2	2
3	27 [b]
5	10
7	7
9	5
$11 \leq h \leq 39$ (odd harmonics only)	3

[a] For some Class C products, other emission limits apply (see 7.4).

[b] The limit is determined based on the assumption of modern lighting technologies having power factors of 0,90 or higher.

Rated power ≥ 5 W and ≤ 25 W



NOTE $I_{p(abs)}$ is the higher absolute value of I_{p+} and I_{p-}.

Lighting equipment having a rated power greater than or equal to 5 W and less than or equal to 25 W shall comply with one of the following three sets of requirements:

– the harmonic currents shall not exceed the power-related limits of Table 3, column 2;

– the third harmonic current, expressed as a percentage of the fundamental current, shall not exceed 86 % and the fifth harmonic current shall not exceed 61 %. In addition, the waveform of the input current shall be such that it reaches the 5 % current threshold before or at 60°, has its peak value before or at 65° and

does not fall below the 5 % current threshold before 90°, referenced to any zero crossing of the fundamental supply voltage. The current threshold is 5 % of the highest absolute peak value that occurs in the measurement window, and the phase angle measurements are made on the cycle that includes this absolute peak value (see Figure 2). Components of current with frequencies above 9 kHz shall not influence this evaluation (a filter similar to the one described in 5.3 of IEC 61000-4-7:2002 and IEC 61000-4-7:2002/AMD1:2008 may be used);

– the *THD* shall not exceed 70 %. The third order harmonic current, expressed as a percentage of the fundamental current, shall not exceed 35 %, the fifth order current shall not exceed 25 %, the seventh order current shall not exceed 30 %, the ninth and eleventh order currents shall not exceed 20 % and the second order current shall not exceed 5 %.

Limits for Class D equipment:

Table 3 – Limits for Class D equipment

Harmonic order h	Maximum permissible harmonic current per watt mA/W	Maximum permissible harmonic current A
3	3,4	2,30
5	1,9	1,14
7	1,0	0,77
9	0,5	0,40
11	0,35	0,33
$13 \leq h \leq 39$ (odd harmonics only)	$\frac{3,85}{h}$	See Table 1

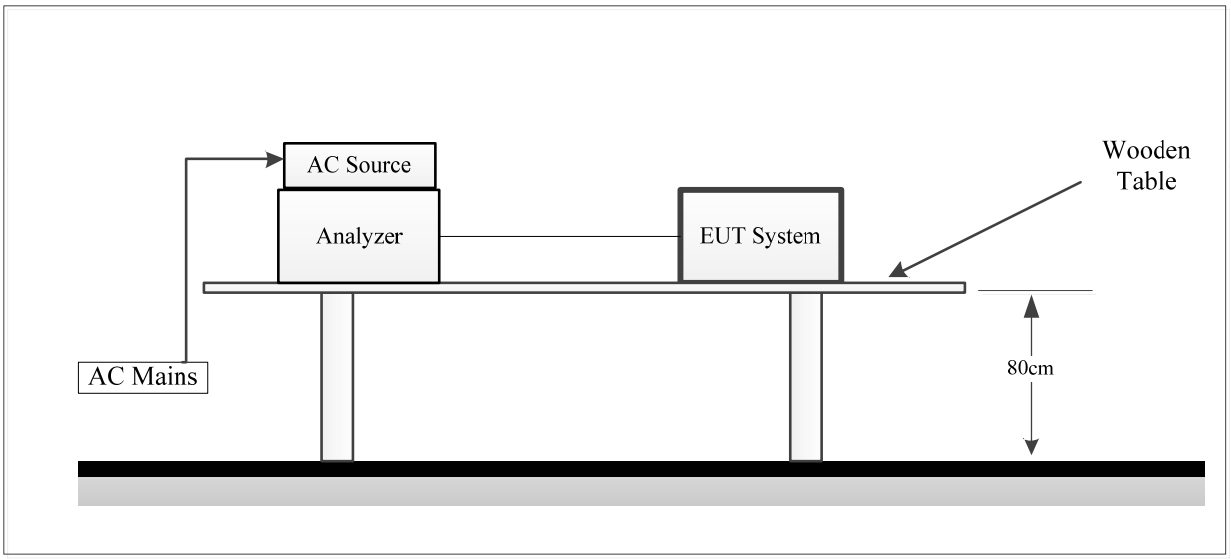
Test Data

Test Result: N/A.

Note: The EUT is powered by DC source.

EN 61000-3-3 VOLTAGE FLUCTUATIONS AND FLICKER

Test System Setup



Test Standard

EN 61000-3-3:2013+A2:2021+AC:2022-01

Test procedure

The EUT was placed on the top of a wooden table 0.8 meters above the ground and operated to produce the most unfavorable sequence of voltage changes under Mode 1conditions.

During the flick measurement, the measure time shall include that part of whole operation cycle in which the EUT produce the most unfavorable sequence of voltage changes. The observation period for short-term flicker indicator is 10 minutes and the observation period for long-term flicker indicator is 2 hours.

Flicker Test Limits

The limits shall be applicable to voltage fluctuations and flicker at the supply terminals of the equipment under test, measured or calculated according to clause 4 under test conditions described in clause 6 and annex A. Tests made to prove compliance with the limits are considered to be type tests.
The following limits apply:
- the value of Pst shall not be greater than 1,0;
- the value of Plt shall not be greater than 0,65;
-Tmax, the accumulated time value of d(t) with a deviation exceeding 3,3 % during a single voltage change at the EUT terminals, shall not exceed 500 ms;
- the relative steady-state voltage change, dc, shall not exceed 3,3 %;
- the maximum relative voltage change dmax, shall not exceed
a) 4 % without additional conditions;
b) 6 % for equipment which is:
Switched manually, or
- switched automatically more frequently than twice per day, and also has either a delayed restart (the delay being not less than a few tens of seconds), or manual restart, after a power supply interruption.

Note: The cycling frequency will be further limited by the Pst and Plt limit. For example: a dmax of 6 % producing a rectangular voltage change characteristic twice per hour will give a Plt of about 0,65.

c) 7 % for equipment which is - attended whilst in use (for example: hair dryers, vacuum cleaners, kitchen equipment such as mixers, garden equipment such as lawn mowers, portable tools such as electric drills), or – switched on automatically, or is intended to be switched on manually, no more than twice per day, and also has either a delayed restart (the delay being not less than a few tens of seconds) or manual restart, after a power supply interruption.

In the case of equipment having several separately controlled circuits in accordance with 6.6, limits b) and c) shall apply only if there is delayed or manual restart after a power supply interruption; for all equipment with automatic switching which is energized immediately on restoration of supply after a power supply interruption, limits a) shall apply; for all equipment with manual switching, limits b) or c) shall apply depending on the rate of switching. Pst and Plt requirements shall not be applied to voltage changes caused by manual switching. The limits shall not be applied to voltage changes associated with emergency switching or emergency interruptions.

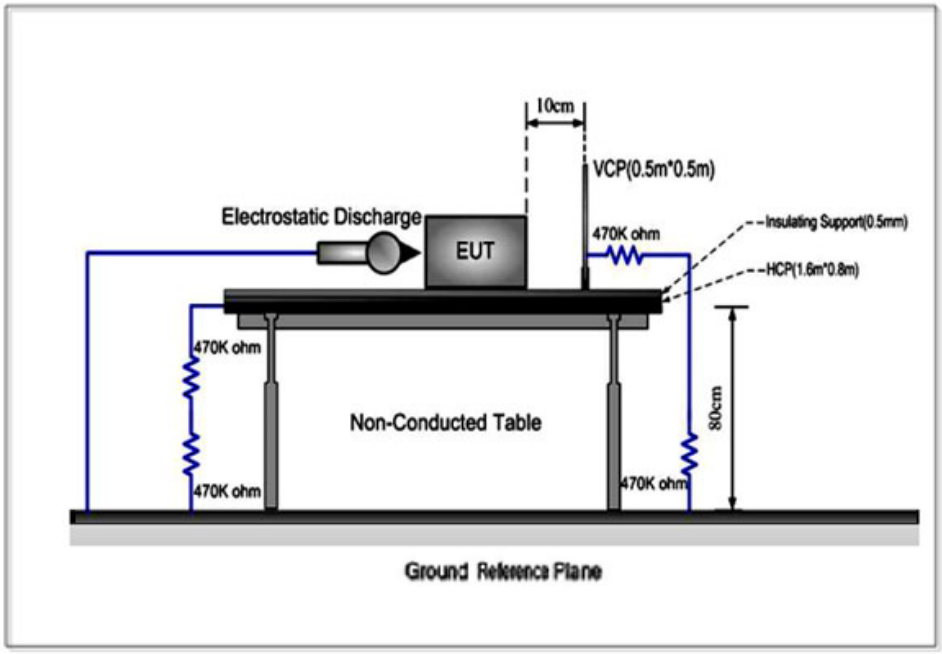
Test Data

Test Result: N/A.

Note: The EUT is powered by DC source.

EN IEC 55014-2 §5.1 ELECTROSTATIC DISCHARGE (IEC 61000-4-2)

Test System Setup



IEC 61000-4-2 specifies that a tabletop EUT shall be placed on a non-conducting table which is 80 centimeters above a ground reference plane and that floor mounted equipment shall be placed on a insulating support approximately 10 centimeters above a ground plane. During the tests, the EUT is positioned over a ground reference plane in conformance with this requirement.

For tabletop equipment, a 1.6m*0.8m metal sheet (HCP) is placed on the table and connected to the ground plane via a metal strap with two 470kΩ resistors in series. The EUT and attached cables are isolated from this metal sheet by 0.5mm thick insulating material. A Vertical Coupling Plane (VCP) grounded on the ground plane through the same configuration as in the HCP is used.

Test Specification

Basic Standard:	EN IEC 55014-2: 2021 / IEC 61000-4-2
Discharge Impedance:	330 ohm / 150 pF
Discharge Voltage:	Air Discharge: ±8 kV (Direct) Contact Discharge: ±4 kV (Direct)
Polarity:	Positive & Negative
Number of Discharge:	Minimum 20 times at each test point
Discharge Mode:	Single Discharge
Discharge Period:	1 second minimum
Performance Criterion:	B

Test Procedure

The basic test procedure was in accordance with IEC 61000-4-2:

Electrostatic discharges were applied only to those points and surfaces of the EUT that are accessible to users during normal operation.

The test was performed with at least ten single discharges on the pre-selected points in the most sensitive polarity.

The time interval between two successive single discharges was at least 1 second.

The ESD generator was held perpendicularly to the surface to which the discharge was applied and the return cable was at least 0.2 meters from the EUT.

Air Discharge: This test is done on a non-conductive surface. The round discharge tip of the discharge electrode shall be approached as fast as possible to touch the EUT. After each discharge, the discharge electrode shall be removed from the EUT. The generator is then re-triggered for a new single discharge and repeated 10 times for each pre-selected test point. This procedure shall be repeated until all the air discharge completed.

Contact Discharge: All the procedure shall be same as Section 8.3.1 of IEC 61000-4-2, except that the tip of the discharge electrode shall touch the EUT before the discharge switch is operated.

Indirect discharge for horizontal coupling plane: At least 20 single discharges shall be applied to the horizontal coupling plane, at points on each side of the EUT. The discharge electrode positions vertically at a distance of 0.1m from the EUT and with the discharge electrode touching the coupling plane.

Indirect discharge for vertical coupling plane: At least 20 single discharges shall be applied to the center of one vertical edge of the coupling plane. The coupling plane, of dimensions 0.5m * 0.5m, is placed parallel to, and positioned at a distance of 0.1m from the EUT. Discharges shall be applied to the coupling plane, with this plane in sufficient different positions that the four faces of the EUT are completely illuminated.

Test Data

Test Model:	SDC-LE-01D-4pin	**Test Mode:**	Mode 1
Test Engineer:	Toby Chen	**Test Date:**	2026-04-30
Test Voltage:	DC 3.3V from serial interface board	**Environment:**	Temp.: 23.8℃ Humi.: 55% Press.: 99.9kPa

Electrostatic Discharge Immunity (Air Discharge)									
Test Levels									
Test Points Location	**-2 kV**	**+2 kV**	**-4 kV**	**+4 kV**	**-8 kV**	**+8 kV**	**-15 kV**	**+15 kV**	**X**
/	/	/	/	/	/	/	/	/	/

Electrostatic Discharge Immunity (Contact Discharge)									
Test Levels									
Test Points Location	**-2 kV**	**+2 kV**	**-4 kV**	**+4 kV**	**-6 kV**	**+6 kV**	**-8 kV**	**+8 kV**	**X**
/	/	/	/	/	/	/	/	/	/

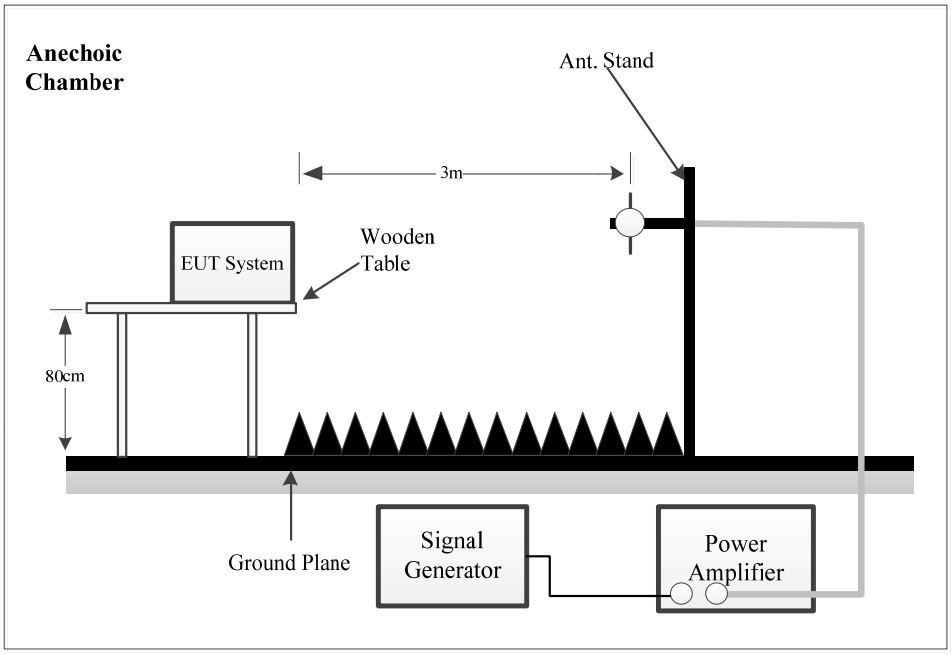
Electrostatic Discharge Immunity (Indirect Contact HCP)									
Test Levels									
Test Points Location	**-2 kV**	**+2 kV**	**-4 kV**	**+4 kV**	**-6 kV**	**+6 kV**	**-8 kV**	**+8 kV**	**X**
Front Side	/	/	A	A	/	/	/	/	/
Back Side	/	/	A	A	/	/	/	/	/
Left Side	/	/	A	A	/	/	/	/	/
Right Side	/	/	A	A	/	/	/	/	/

Electrostatic Discharge Immunity (Indirect Contact VCP)									
Test Levels									
Test Points Location	**-2 kV**	**+2 kV**	**-4 kV**	**+4 kV**	**-6 kV**	**+6 kV**	**-8 kV**	**+8 kV**	**X**
Front Side	/	/	A	A	/	/	/	/	/
Back Side	/	/	A	A	/	/	/	/	/
Left Side	/	/	A	A	/	/	/	/	/
Right Side	/	/	A	A	/	/	/	/	/

Note: “A” stands for, during the test, the apparatus shall continue to operate as intended during the test. No degradation of performance or loss of function is allowed below a performance level (or permissible loss of performance) specified by the manufacturer, when the apparatus is used as intended.

EN IEC 55014-2 §5.5 RADIO FREQUENCY ELECTROMAGNETIC FIELDS (IEC 61000-4-3)

Test System Setup



Test specification

Basic Standard:	EN IEC 55014-2:2021 / IEC 61000-4-3
Frequency Range:	80 MHz - 1000 MHz
Field Strength:	3 V/m
Modulation:	1kHz Sine Wave, 80%, AM Modulation
Frequency Step:	1 % of fundamental
Polarity of Antenna:	Horizontal and Vertical
Antenna Height:	1.5m
Dwell Time:	3seconds
Performance Criterion:	A

Test Procedure

The testing was performed in a fully-anechoic chamber.

The frequency range is swept from 80 MHz to 1000 MHz, with the signal 80% amplitude modulated with a 1kHz sinewave.

The dwell time at each frequency shall not be less than the time necessary for the EUT to be exercised and to respond, but shall in no case be less than 0.5s.

The field strength level was 3V/m.

The EUT and its simulators are placed on a turn table which is 0.8 meter above the ground. The EUT is set 3 meters away from the transmitting antenna which is mounted on an antenna tower. Both horizontal and vertical polarizations of the antenna are set on test. Each of the four sides of EUT must be faced this

transmitting antenna and measured individually.

In order to judge the EUT performance, a CCD camera is used to monitor the EUT.

Test Data

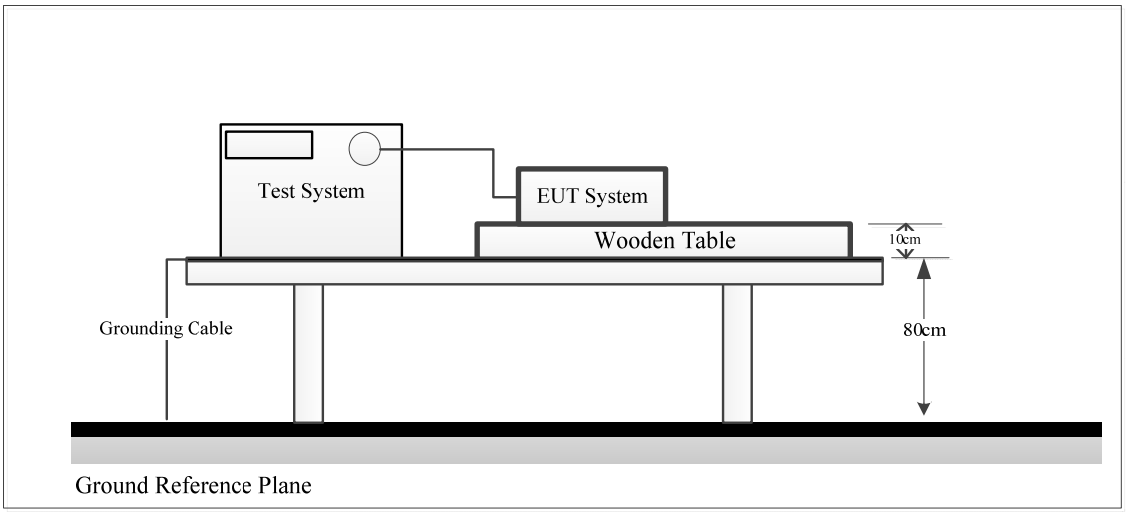
Test Model:	SDC-LE-01D-4pin	**Test Mode:**	Mode 1
Test Engineer:	Toby Chen	**Test Date:**	2026-04-30
Test Voltage:	DC 3.3V from serial interface board	**Environment:**	Temp.: 24.2℃ Humi.: 56% Press.: 99.9kPa

Frequency [MHz]	**Level [V/m]**	**Modulation**	**Antenna Polarization**	**EUT Face**	**Observations (Performance Criterion)**
80 - 1000	3	1 kHz, 80 % Amp. Mod, 1 % increment, dwell time=3seconds	V	Front	A
			H		A
			V	Rear	A
			H		A
			V	Left	A
			H		A
			V	Right	A
			H		A
			V	Top	A
			H		A
			V	Bottom	A
			H		A

Note: “A” stands for, during the test, the apparatus shall continue to operate as intended during the test. No degradation of performance or loss of function is allowed below a performance level (or permissible loss of performance) specified by the manufacturer, when the apparatus is used as intended.

EN IEC 55014-2 §5.2 ELECTRIC FAST TRANSIENTS (IEC 61000-4-4)

Test System Setup



Test specification

Basic Standard:	EN IEC 55014-2:2021 / IEC 61000-4-4
Test Voltage:	Input and output AC power ports: 1 kV Input and output DC power ports: 0.5 kV Signal ports, control ports and wired network ports: 0.5 kV
Polarity:	Positive & Negative
Impulse Frequency:	5 kHz
Impulse Wave Shape :	5/50 ns
Burst Duration:	15 ms
Burst Period:	300 ms
Test Duration:	2 min.
Performance Criterion:	B

Test Procedure

The EUT was arranged for Power Line Coupling and for I/O Line Coupling through a capacitive clamp, where applicable. (Note: The I/O coupling test using a capacitive clamp is performed on the I/O interface cables that are longer in length than 3 meters.) A metal ground plane is connected to the earth by a ground rod. The ground rod is connected to the test facility's electrical earth.

Both positive and negative polarity discharges were applied.

The length of the "hot wire" from the coaxial output of the EFT generator to the terminals on the EUT should not exceed 0.5 meter ± 0.05 meter.

The duration time of each test sequential was 2 minute.

The transient/burst waveform was in accordance with IEC 61000-4-4, 5/50ns.

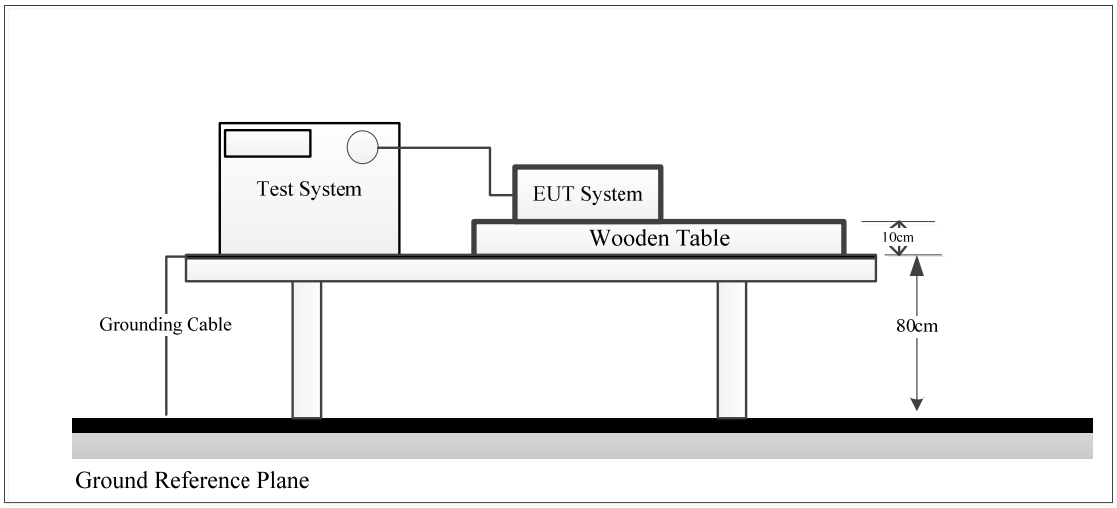
Test Data

Test Result: N/A.

Note: The EUT is powered by DC source.

EN IEC 55014-2 §5.6 SURGES (IEC 61000-4-5)

Test System Setup



Test Specification

Basic Standard:	EN IEC 55014-2:2021 / IEC 61000-4-5
Wave-Shape:	Combination Wave 1.2/50 us Open Circuit Voltage 8 /20 us Short Circuit Current
Test Voltage:	Power Line : Line - line: 1kV, Line - earth: 2kV.
Generator Source Impedance:	2 ohm between networks 12 ohm between network and ground
Polarity:	Positive/Negative
Phase Angle:	5 positive for 90°, 5 negative for 270°
Pulse Repetition Rate:	1 time / 60 Sec.
Performance Criterion:	B

Test Procedure

For input a.c. power ports, provide a 1.2/50μs voltage surge (at open-circuit condition) and a 8/20 μs current surge into a short circuit.

For telecommunication port, provide a 10/700μs voltage surge (at open-circuit condition) and a 5/320 μs current surge into a short circuit.

At least 5 positive and 5 negative (polarity) tests with a maximum 1/min repetition rate are conducted during test.

Different phase angles are done individually.

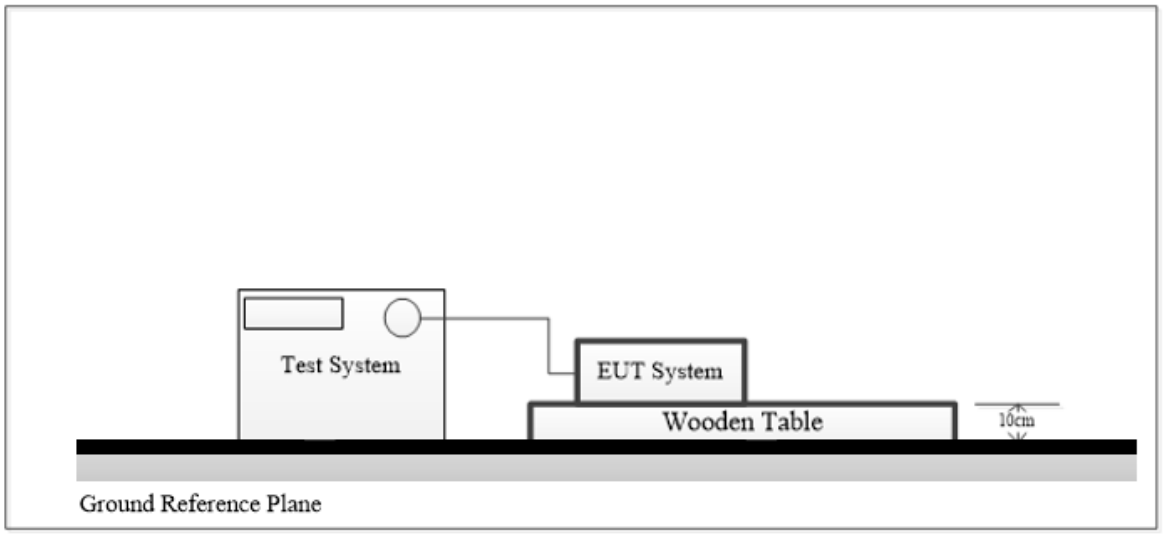
Record the EUT operating situation during compliance test and decide the EUT immunity criterion for above each test.

Test Data

Test Result: N/A.
Note: The EUT is powered by DC source.

EN IEC 55014-2 §5.4 INJECTED CURRENTS (IEC 61000-4-6)

Test System Setup



Test specification

Basic Standard:	EN IEC 55014-2:2021 / IEC 61000-4-6
Frequency Range:	0.15 MHz - 80 MHz
Field Strength:	Input and output AC power ports: 3 $V_{r.m.s}$ Input and output DC power ports: 1 $V_{r.m.s}$ Signal ports, control ports, wired network ports: 1 $V_{r.m.s}$
Modulation:	1 kHz Sine Wave, 80%, AM Modulation
Frequency Step:	1 % of fundamental
Dwell Time:	1 second
Performance Criterion:	A

Test Procedure

The EUT was placed on an insulating support of 0.1m height above a ground reference Plane, arranged and connected to satisfy its functional requirement.

All relevant cables were provided with the appropriate coupling and decoupling devices at a distance between 0.1m and 0.3m from the projected geometry of the EUT on an insulating support of 0.03m height above the ground reference plane.

The test shall be performed with the test generator connected to each of the coupling and decoupling devices in turn, while the other non-excited RF input ports of the coupling devices are terminated by a 50-ohm load resistor.

The frequency range is swept from 150 kHz to 80 MHz, using the signal level established during the setting process and with a disturbance signal of 80 % amplitude. The signal is modulated with a 1 kHz sine wave, pausing to adjust the RF signal level or the switch coupling devices as necessary. Where the frequency is swept incrementally, the step size shall not exceed 1 % of the preceding frequency value.

The dwell time of the amplitude modulated carrier at each frequency shall not be less than the time necessary for the EUT to be exercised and to respond, but shall in no case be less than 0.5 s. The sensitive frequencies (e.g. clock frequencies) shall be analyzed separately.

Attempts should be made to fully exercise the EUT during testing, and to fully interrogate all exercise modes selected for susceptibility.

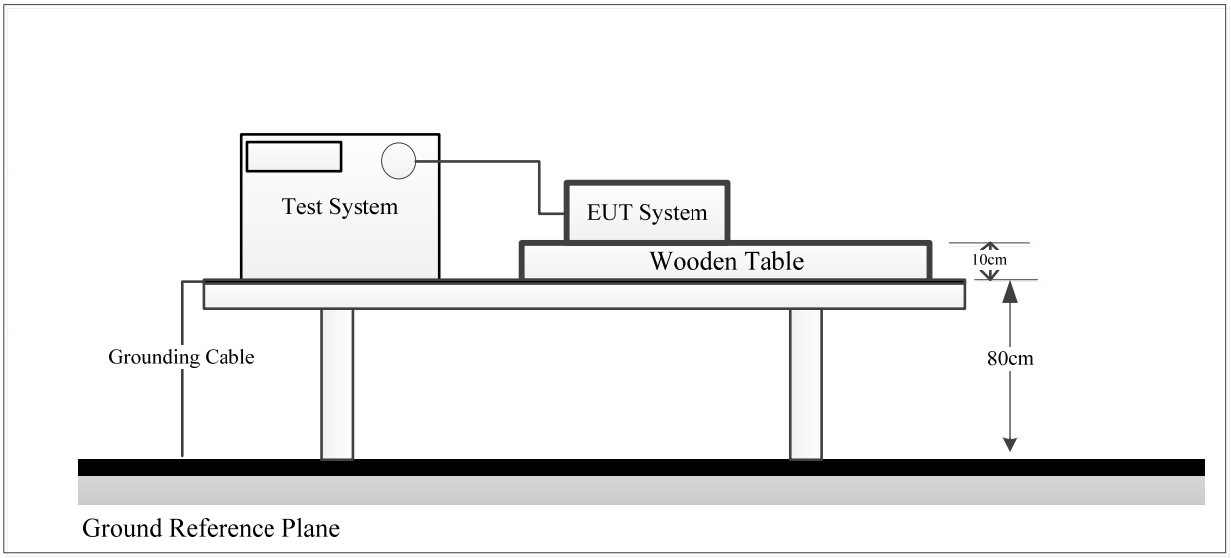
Test Data

Test Result: N/A.

Note: The EUT is powered by DC source.

EN IEC 55014-2 §5.7 VOLTAGE DIPS AND INTERRUPTIONS (IEC 61000-4-11)

Test System Setup



Test specification

Basic Standard:	EN IEC 55014-2:2021 / IEC 61000-4-11			
Test Duration Time:	Minimum three test events in sequence			
Interval between Event:	Minimum ten seconds			
Phase Angle:	0°/180°			
Numbers of dropout:	3 times			
Performance Criterion:	**Test Level**	**Cycle**	**Phase Angle**	**Performance criterion**
	0 %	0.5	0º/180º	C
	40 %	10 for 50Hz 12 for 60Hz	0º/180º	C
	70 %	25 for 50Hz 30 for 60Hz	0º/180º	C

Test Procedure

The EUT shall be tested for each selected combination of test levels and duration with a sequence of tree dips/interruptions with intervals of 10 s minimum (between each test event). Each representative mode of operation shall be tested. Abrupt changes in supply voltage shall occur at zero crossings of the voltage waveform.

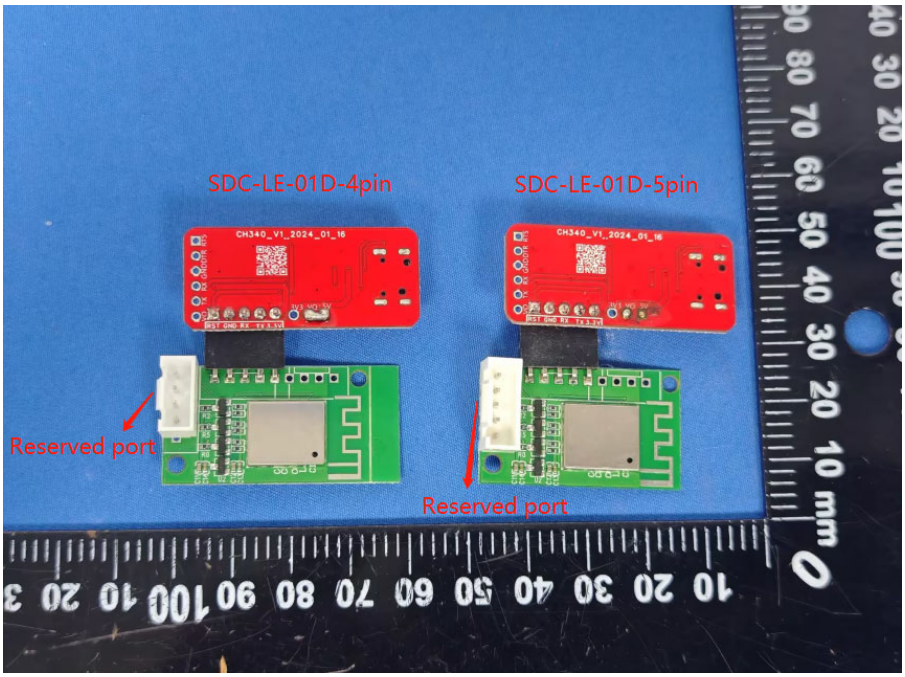
Test Data

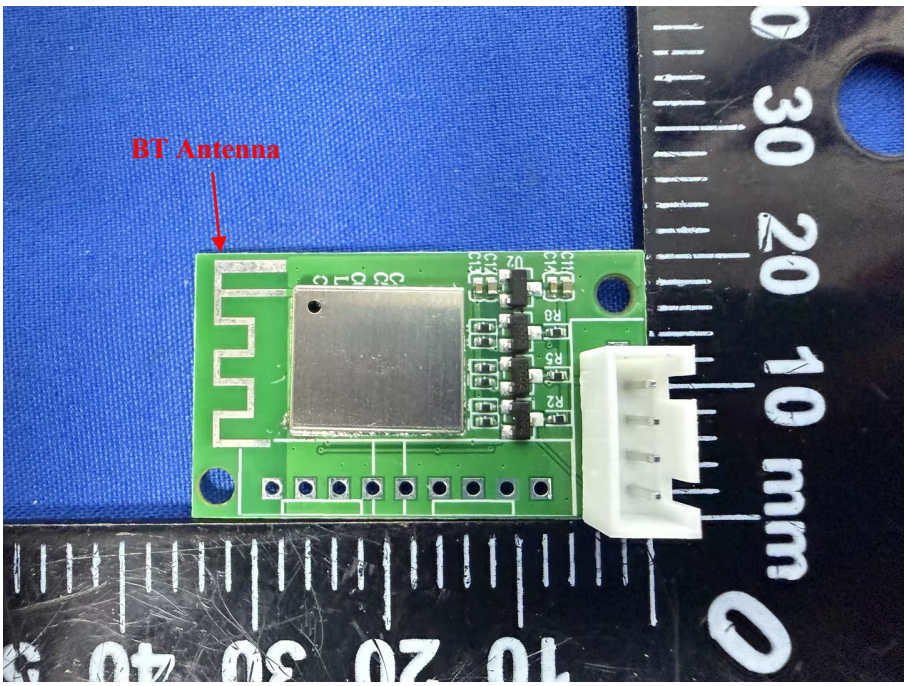
Test Result: N/A.

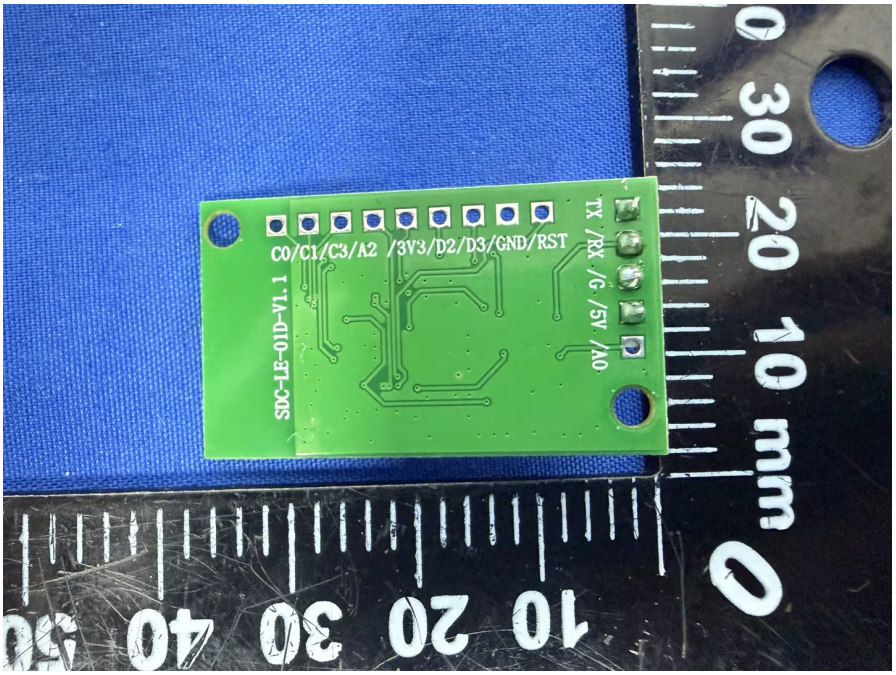
Note: The EUT is powered by DC source.

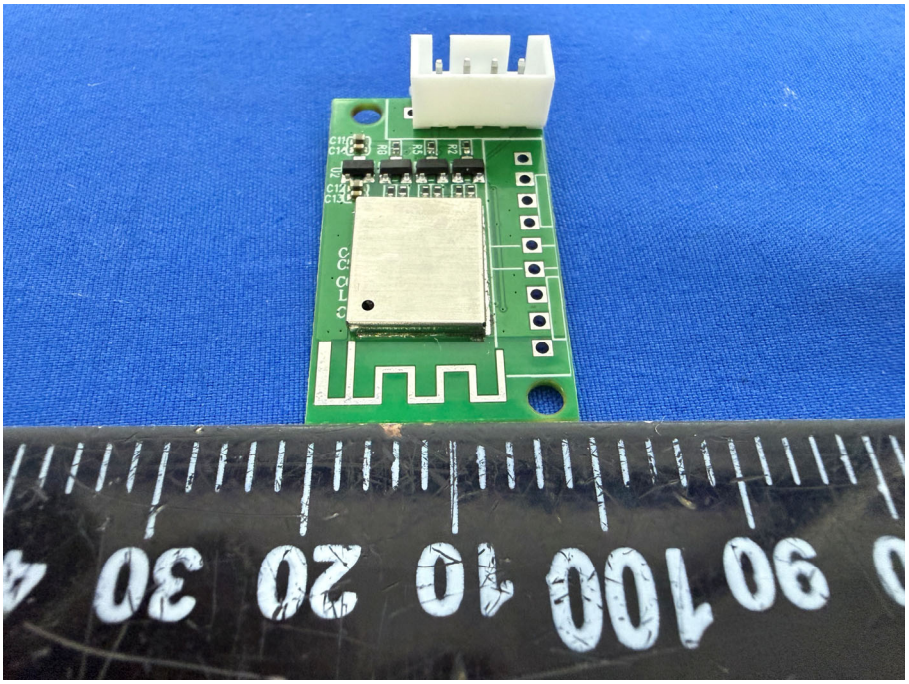
EXHIBIT A - EUT PHOTOGRAPHS

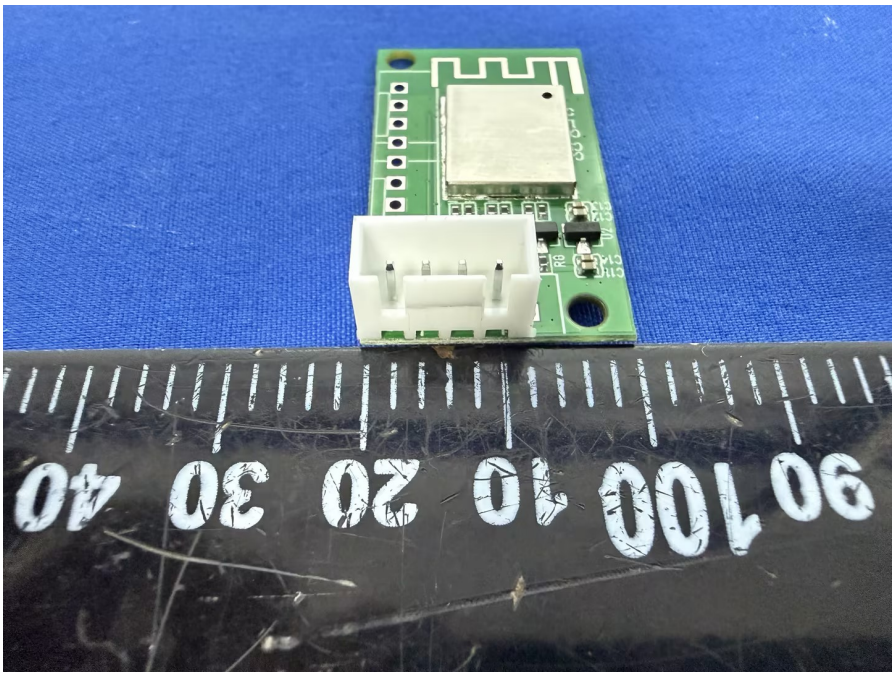
External Photos

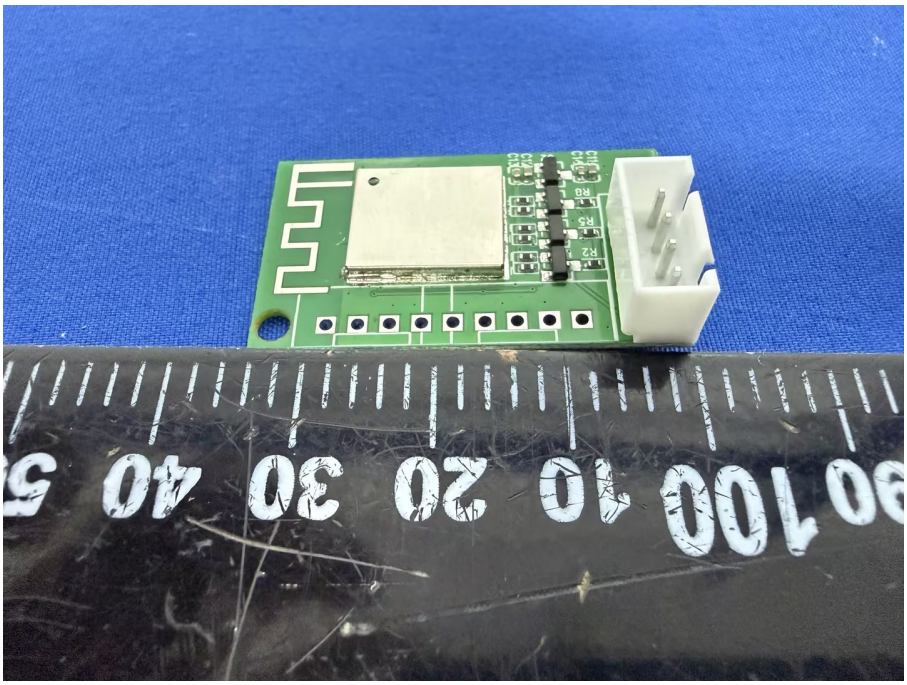


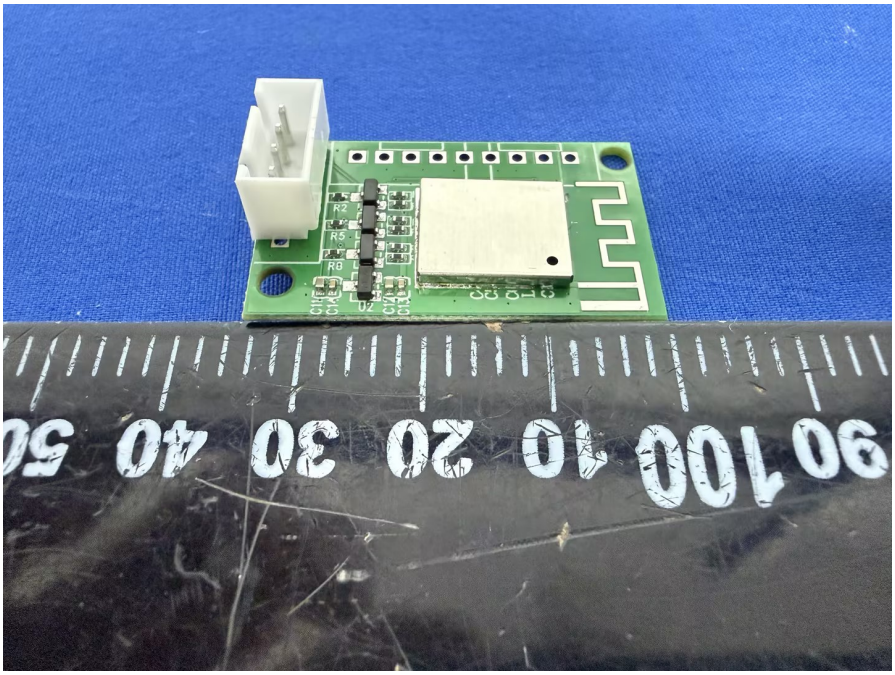



C0/C1/C3/A2 /3V3/D2/D3/GND/RST
TX /RX /G /5V /A0
SDC-LE-01D-V1.1
0 30 20 10 mm 0
10 20 30 40


90 100 10 20 30







Internal Photos

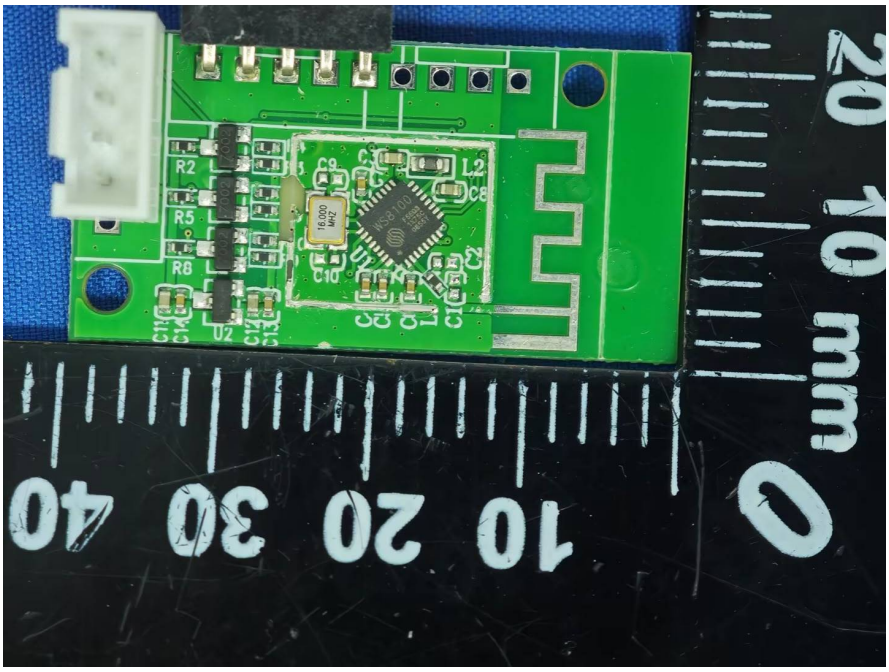
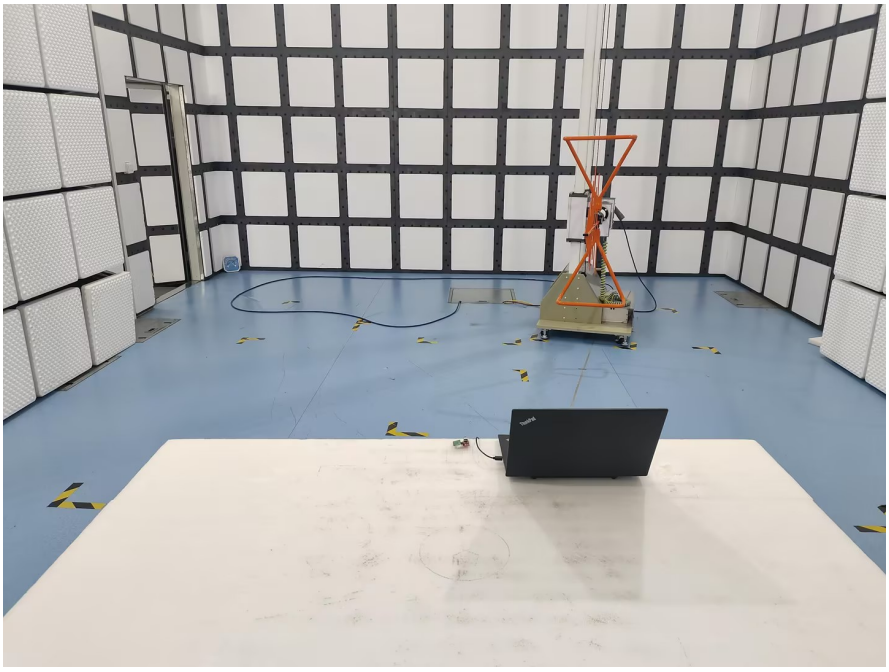
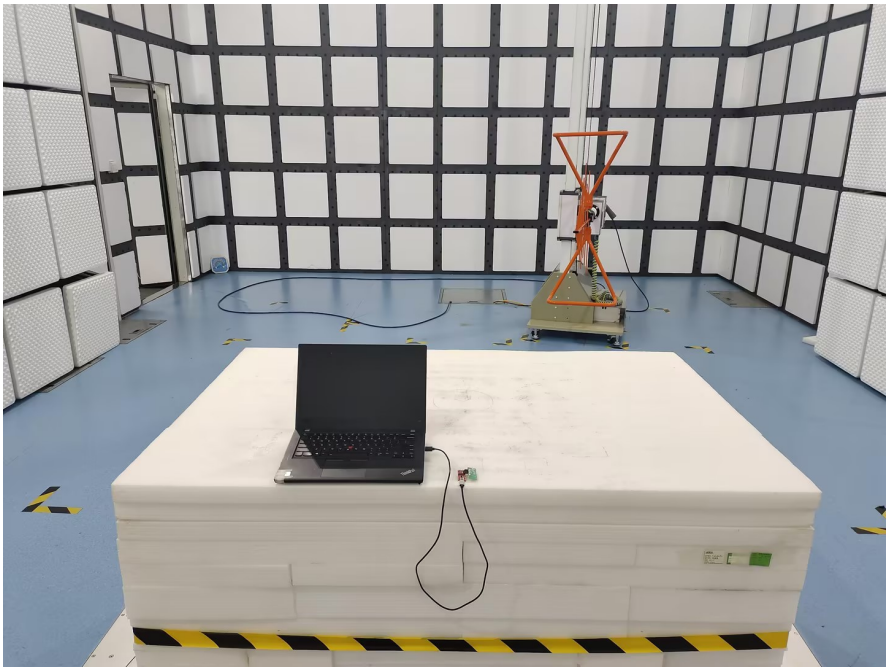


EXHIBIT B - TEST SETUP PHOTOGRAPHS

RE - Front View(30MHz-1000MHz)



RE - Back View(30MHz-1000MHz)



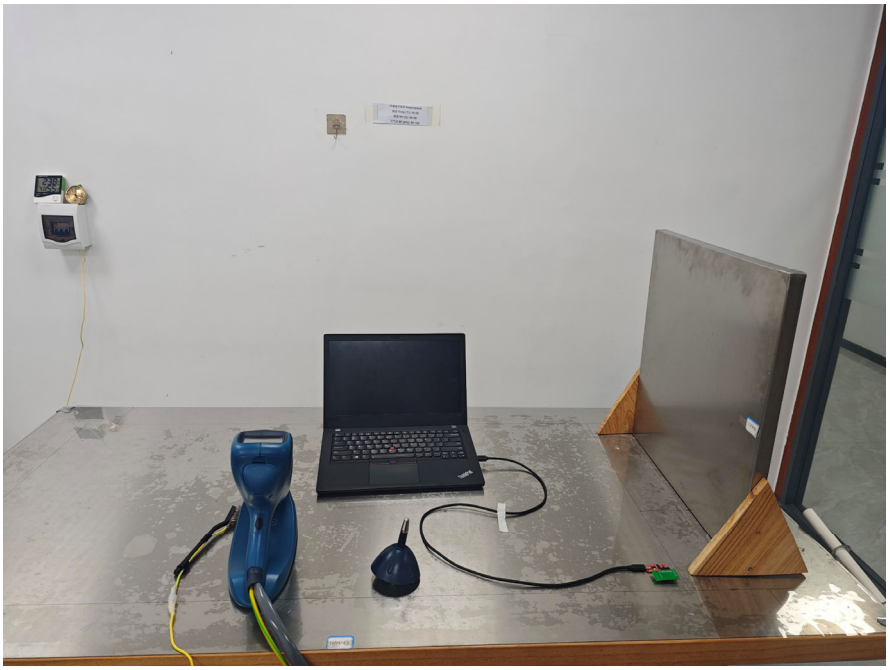
RS - Front View (Below 1GHz)



RS - Back View (Below 1GHz)



Electrostatic Discharge



Declarations

1. Bay Area Compliance Laboratories Corp. (Fujian) is not responsible for authenticity of any information provided by the applicant. Information from the applicant that may affect test results are marked with an asterisk "★".

2. Unless otherwise stated, the results shown in this test report refer only to the sample(s) tested.

3. Unless required by the rule provided by the applicant or product regulations, then decision rule in this report did not consider the uncertainty.

4. The extended uncertainty given in this report is obtained by combining the standard uncertainty times the coverage factor k=2 with the 95% confidence interval.

5. This report cannot be reproduced except in full, without prior written approval of Bay Area Compliance Laboratories Corp. (Fujian).

6. This report is valid only with a valid digital signature. The digital signature may be available only under the adobe software above version 7.0.

PRODUCT SIMILARITY DECLARATION LETTER

Quanzhou STAYTRUE Electronic Technology Co., Ltd
Unit 1A-2, 1st Floor, Building 1, Fengshu Logistics Park, No. 920, Binhu South Road, Taiwan Investment Zone, Quanzhou, Fujian Province, China

Declaration of Model Difference

To Whom It May Concern,

We Quanzhou STAYTRUE Electronic Technology Co., Ltd hereby declare that there are some differences between series models and tested model(s). Details are as below:

Products Description	Name:	Module SDC-LE-01D	
	Brand:	SDC-LE-01D	
	Manufacturer:	Quanzhou STAYTRUE Electronic Technology Co., Ltd	
	Project No.:	2607S04032E-EM、2607S04032E-RF	
Differences Description			
Tested Model(s)	Series Models	Differences Items	Details
SDC-LE-01D-4pin	SDC-LE-01D-5pin	Model Name, Number of pins	All are the same except model name, number of pins (SDC-LE-01D-4pin: 4pins SDC-LE-01D-5pin: 5pins)

Note: Tested Model(s) mean the models have been tested by Bay Area Compliance Laboratories Corp. (Fujian).

Except for the differences in above table, we declare the products are identical in every other way. We guarantee all the information provided above is true, and notice that we'll bear all the consequences caused by any false information or concealing.

Best Regards,

Signature: Guiping Lin

Print Name: Guiping Lin
Title: Manager

*******END OF REPORT*******